



GCSE BIOLOGY

ULTIMATE REVISION GUIDE

Botany | Health & Disease | Evolution & Inheritance

Complete • Detailed • Exam-focused • Beautifully structured



includes plant cells, photosynthesis, reproduction, pathogens, immunity, diet, inheritance, natural selection, fossils and practical skills

How to Use This Guide

This PDF is designed for your Biology test on Monday. It includes the three areas you gave: **Botany**, **Health and Disease**, and **Evolution**. It is detailed but stays focused on the content you actually revised.

Best revision method

- Read one section quickly.
- Close the guide and recall the key facts from memory.
- Use the exam traps and definitions to check precision.
- Tomorrow, test yourself one question at a time.

Contents

1. Botany and plant cells
2. Specialised plant cells and transport
3. Leaf structure, stomata and transpiration
4. Photosynthesis and the pondweed practical
5. Xerophytes, carnivorous plants and classification
6. Flower structure, pollination, fertilisation and seed dispersal
7. Health and disease
8. Pathogens: bacteria, viruses, fungi and protists
9. Case studies: Salmonella, Covid-19, Ebola and Ringworm
10. Body defences, white blood cells, medicines and vaccination
11. Balanced diet, deficiencies and inherited diseases
12. Evolution, natural selection, Darwin, moths and fossils
13. Data skills, definitions and exam traps

SECTION 1: BOTANY

Plant cells, photosynthesis, transport, reproduction and adaptations

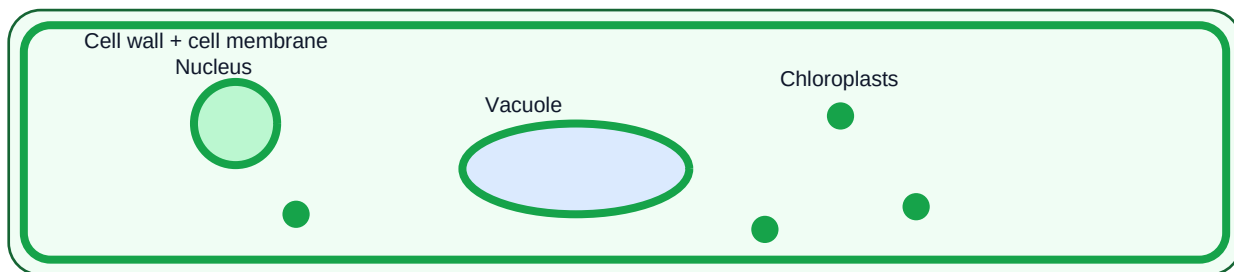
1. What is Botany?

Botany is the study of plants. Plants are usually multicellular, eukaryotic organisms that make glucose by photosynthesis. They are producers in food chains because they make their own food rather than eating other organisms.

Plants must be able to

- photosynthesise to make glucose
- respire to release energy
- absorb water and mineral ions
- transport substances using xylem and phloem
- exchange gases through stomata
- reproduce sexually or asexually
- adapt to their environment

2. Plant Cell Structure



Structure	Detailed function	High-mark wording
Cell wall	Made of cellulose. Strengthens, supports and keeps the cell rigid. Helps prevent bursting when water enters.	The cell wall strengthens and supports the plant cell.
Cell membrane	Controls movement of substances into and out of the cell. It is selectively/partially permeable.	The cell membrane controls movement of substances in and out.
Cytoplasm	Jelly-like substance where many chemical reactions happen.	The cytoplasm is the site of chemical reactions.
Nucleus	Contains DNA/genetic material and controls cell activities.	The nucleus controls the cell and contains DNA.
Chloroplast	Organelle where photosynthesis takes place. Contains chlorophyll.	Chloroplasts are the site of photosynthesis.
Chlorophyll	Green pigment that absorbs light energy for photosynthesis.	Chlorophyll absorbs light energy.
Vacuole	Contains cell sap and maintains turgor pressure, keeping the plant cell firm.	The vacuole contains cell sap and maintains turgor pressure.

Plant cells vs animal cells

- Plant cells have a cell wall, chloroplasts and a large permanent vacuole.
- Animal cells do not have these.
- Both plant and animal cells have a nucleus, cytoplasm and cell membrane.

3. Specialised Plant Cells

A specialised cell is adapted to carry out a particular function. In your test, the most important specialised plant cells are palisade cells, root hair cells, xylem vessels, phloem cells, guard cells and pollen cells.

Palisade Cells

Feature	Explanation
Location	Near the top of the leaf in the palisade mesophyll layer.
Function	Main site of photosynthesis.
Many chloroplasts	Absorb maximum light energy for photosynthesis.
Near top of leaf	Receives the most light.
Packed closely together	Allows lots of photosynthetic cells to fit near the light source.

Root Hair Cells

Adaptation	How it helps
Long hair-like projection	Creates a large surface area for absorbing water and mineral ions from soil.
Thin wall	Shorter distance for water movement by osmosis.
Large vacuole	Helps maintain a water potential gradient so water enters.
No chloroplasts	They are underground where little/no light reaches, so photosynthesis cannot occur.

Osmosis - must know

- Osmosis is the movement of water through a partially permeable membrane from a high water concentration to a low water concentration.
- Water enters root hair cells by osmosis.
- Mineral ions can be absorbed by active transport if moving against a concentration gradient.

Xylem and Phloem

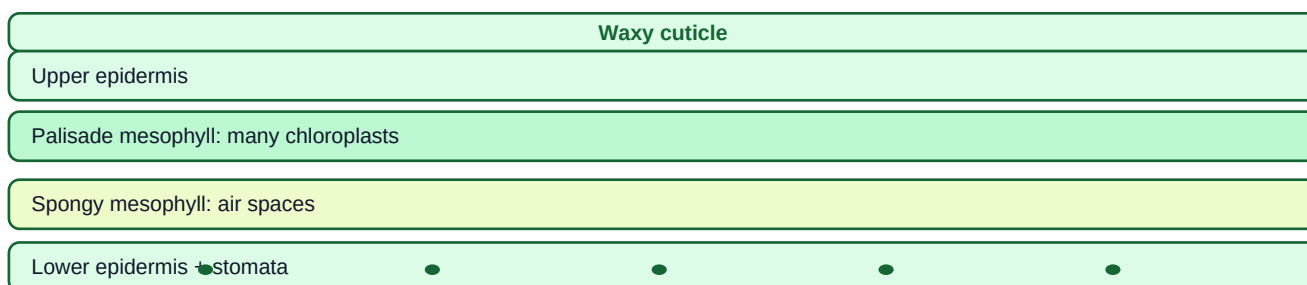
Feature	Xylem	Phloem
What it transports	Water and mineral ions	Dissolved sugars such as sucrose
Direction	One way: roots to leaves	Both directions
Cell type	Dead cells	Living cells
Structural adaptations	Hollow tubes, no end walls, lignin strengthening	Sieve plates and companion cells
Process name	Transpiration stream	Translocation

Guard Cells and Pollen Cells

Guard cells open and close stomata. When guard cells are turgid, stomata open; when they lose water, stomata close. This controls gas exchange and water loss.

Pollen grains contain the male nucleus/gamete. Their job is to deliver the male nucleus to the ovule during fertilisation.

4. Leaf Structure



Leaf part	Function
Waxy cuticle	Waterproof layer that reduces water loss by evaporation.
Upper epidermis	Transparent protective layer that lets light through.
Palisade mesophyll	Main photosynthesis layer; contains many chloroplasts.
Spongy mesophyll	Contains air spaces for diffusion of gases.
Lower epidermis	Contains stomata.
Veins	Contain xylem and phloem for transport.

5. Stomata, Gas Exchange and Transpiration

Stomata are tiny pores mainly found on the underside of leaves. They allow gas exchange.

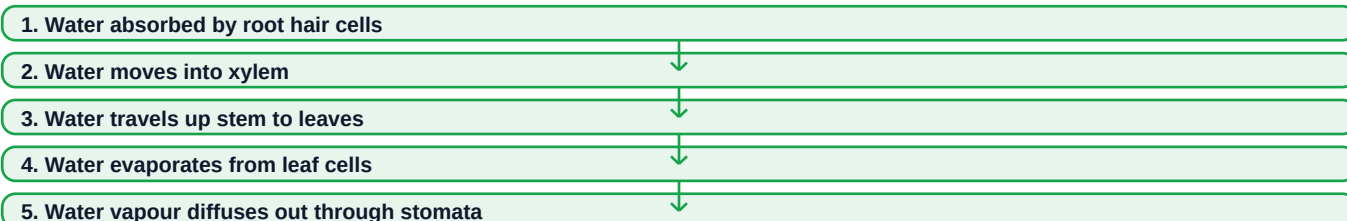
Substance	Direction through stomata	Why
Carbon dioxide	Enters leaf	Used in photosynthesis.
Oxygen	Leaves leaf	Produced in photosynthesis.
Water vapour	Leaves leaf	Lost during transpiration.

Diffusion vs osmosis

- Diffusion moves particles/gases from high concentration to low concentration.
- Osmosis is only the movement of water through a partially permeable membrane.
- Gas exchange through stomata happens by diffusion, not osmosis.

Transpiration

Transpiration is the loss of water vapour from a plant through the stomata. It helps pull water up the plant through xylem and can help cool the plant.



6. Photosynthesis

PHOTOSYNTHESIS

Carbon dioxide + water → glucose + oxygen



Photosynthesis is the process where plants convert light energy into chemical energy stored in glucose.

Needed for photosynthesis	Role
Light energy	Provides energy for the reaction.
Chlorophyll	Absorbs light energy.
Carbon dioxide	Reactant that enters through stomata.
Water	Reactant absorbed by roots and transported in xylem.

Uses of Glucose

Use	Explanation
Respiration	Glucose is broken down to release energy.
Stored as starch	Starch is an insoluble storage molecule.
Cellulose	Used to make cell walls.
Proteins	Glucose combines with nitrates to make amino acids/proteins.
Fats and oils	Converted into lipids for storage and growth.

Limiting Factors

A limiting factor is something that restricts the rate of photosynthesis. The main limiting factors are light intensity, carbon dioxide concentration, temperature and chlorophyll amount. Temperature matters because photosynthesis uses enzymes: too cold means slow reactions, too hot can denature enzymes.

7. Pondweed Practical

The pondweed practical investigates how light intensity affects the rate of photosynthesis.

Variable type	In this practical	Meaning
Independent variable	Distance from lamp / light intensity	What you change.
Dependent variable	Number of oxygen bubbles per minute	What you measure.
Control variables	Temperature, pondweed species, water volume, time counted, carbon dioxide concentration	What you keep the same.

1. Place pondweed in water

2. Set lamp at measured distance

3. Count oxygen bubbles per minute

4. Repeat at different distances

5. Compare rate of photosynthesis

Expected conclusion

- As the lamp gets closer, light intensity increases.
- The rate of photosynthesis increases.
- More oxygen bubbles are produced per minute.

8. Xerophytes and Carnivorous Plants

Xerophytes

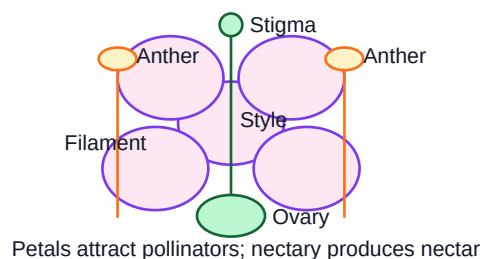
Xerophytes are plants adapted to dry environments. Cacti are common examples.

Adaptation	How it helps survival
Thick waxy cuticle	Reduces water loss by evaporation.
Spines instead of large leaves	Reduce surface area, reducing transpiration; also protect against animals.
Sunken stomata	Trap moist air, reducing the water concentration gradient and transpiration.
Fewer stomata	Less water vapour can escape.
Water storage tissue	Stores water for dry periods.
Deep or wide roots	Help absorb water from a larger area/deeper ground.

Carnivorous Plants

Examples include Venus flytraps and pitcher plants. Carnivorous plants often live in nutrient-poor, nitrogen-poor soil, so they catch insects to obtain nitrogen and minerals. They still photosynthesise for glucose; insects are mainly for minerals, not their main energy source.

9. Plant Reproduction and Flower Structure



Flower part	Type	Function
Anther	Male	Produces pollen grains containing male nuclei/gametes.
Filament	Male	Supports the anther and holds it up.
Stigma	Female	Sticky structure that catches pollen.
Style	Female	Connects stigma to ovary; pollen tube grows down it.
Ovary	Female	Contains ovules; becomes fruit after fertilisation.
Ovule	Female	Contains female nucleus/gamete; becomes seed after fertilisation.
Petals	Other	Attract pollinators with colour/scent.

Flower part	Type	Function
Nectary	Other	Produces nectar to attract insects.
Sepals	Other	Protect the flower bud.

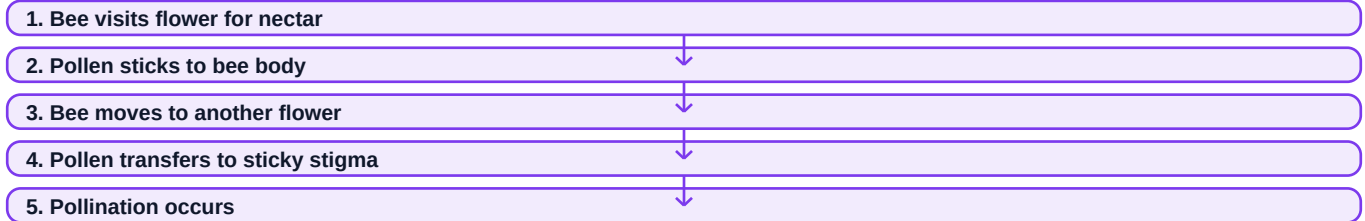
10. Pollination, Fertilisation and Seed Dispersal

Pollination

Pollination is the transfer of pollen from the anther to the stigma.

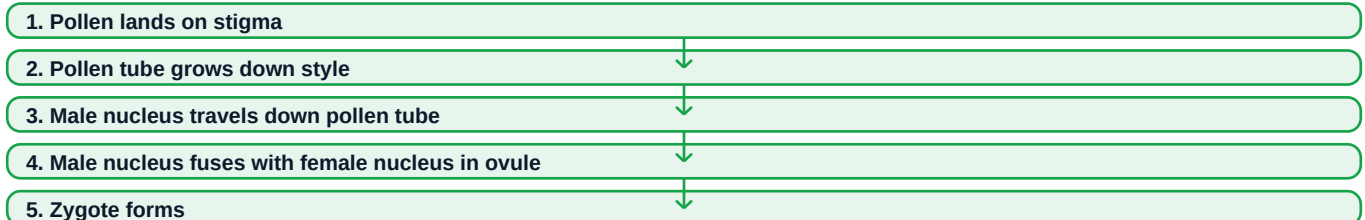
Insect-pollinated flowers	Wind-pollinated flowers
Bright petals, scent, nectar, sticky pollen, sticky stigma	Dull/small petals, no nectar, exposed anthers, feathery stigmas, light pollen

Role of Bees



Fertilisation

Fertilisation is the fusion of the male nucleus and female nucleus.



Before fertilisation	After fertilisation
Ovule	Seed
Ovary	Fruit
Fertilised egg cell	Zygote/embryo

Sexual vs Asexual Reproduction

Feature	Sexual reproduction	Asexual reproduction
Parents	Usually two	One
Gametes	Yes	No
Fertilisation	Yes	No
Offspring	Genetically varied	Clones
Advantage	Variation helps survival in changing environments	Fast; no pollinator/mate needed

Feature	Sexual reproduction	Asexual reproduction
Disadvantage	Slower and uses energy	No variation; vulnerable to disease/environment change

Seed Dispersal

Seed dispersal reduces competition between seedlings and the parent plant for light, water, space and mineral ions.

Method	Description	Example
Wind	Light seeds are carried by air.	Dandelion
Water	Seeds float and are carried away.	Coconut
Animals	Seeds stick to fur or are eaten and excreted.	Hooked seeds/berries
Explosion	Seed pods burst and scatter seeds.	Pea pods
Gravity	Heavy fruits fall and may roll away.	Apples

11. Plant Classification

Group	Key features	Examples
Flowering plants / Angiosperms	Produce flowers; seeds enclosed in fruits.	Roses, grasses, fruit trees
Gymnosperms	Non-flowering seed plants; exposed/naked seeds.	Pine trees, conifers
Ferns	Non-flowering plants that reproduce using spores.	Ferns
Mosses	Small non-flowering plants, often in damp areas.	Moss

SECTION 2: HEALTH & DISEASE

Pathogens, case studies, immunity, medicines, diet and inherited disease

12. Health, Disease and Pathogens

Health is a state of physical and mental wellbeing. A **disease** is a condition that affects normal body function.

Term	Meaning	Examples
Communicable disease	Can spread between organisms; usually caused by pathogens.	Covid-19, Salmonella, Ebola, Ringworm
Non-communicable disease	Cannot spread between organisms; not caused by pathogens.	Inherited diseases, deficiency diseases
Pathogen	A microorganism that causes disease.	Bacteria, viruses, fungi, protists

13. Types of Pathogen

Pathogen type	Key features	Disease examples
Bacteria	Unicellular prokaryotes; no nucleus; circular DNA.	Salmonella, cholera, tuberculosis, strep throat
Viruses	Not made of cells; capsid + genetic material; reproduce only inside host cells.	Covid-19, Ebola
Fungi	Can cause skin infections; often spread by contact.	Ringworm, athlete's foot
Protists	Single-celled eukaryotes; some cause disease.	Malaria is a common GCSE example, if taught

14. Bacteria in Detail

Bacteria are unicellular prokaryotes. Prokaryotic means they do not have a nucleus.

Bacterial structure	Function/detail
Cell wall	Supports and protects the bacterial cell.
Cell membrane	Controls movement in and out.
Cytoplasm	Chemical reactions happen here.
Circular DNA	Genetic material, loose in the cytoplasm.
No nucleus	DNA is not enclosed in a nucleus.
Flagellum	Some bacteria have one for movement.

Salmonella Case Study

Category	Details
Type	Bacterial disease.
Spread	Contaminated food, undercooked food, poor hygiene, cross-contamination.

Category	Details
Symptoms	Fever, diarrhoea, vomiting, stomach cramps.
Prevention	Cook food properly, wash hands, clean surfaces, store food safely.
Treatment	Fluids/rest; antibiotics may be used in serious bacterial infections.

15. Viruses in Detail



Capsid = protein coat
No nucleus, cytoplasm or cell membrane

Viruses are not considered fully living because they do not carry out life processes independently and cannot reproduce without infecting a host cell.

MRS GREN - life processes

- Movement
- Respiration
- Sensitivity
- Growth
- Reproduction
- Excretion
- Nutrition
- Viruses do not perform these independently.

Virus Structure

Part	Function
Capsid	Protein coat that protects genetic material.
Genetic material	DNA or RNA with instructions for making new viruses.

Viral Reproduction

1. Virus attaches to host cell
2. Virus injects genetic material
3. Host cell copies viral material
4. New viruses are assembled
5. Host cell bursts/releases viruses
6. Viruses infect more cells

16. Fungi and Case Studies

Fungi

Fungi can cause diseases, especially skin infections. Examples include ringworm and athlete's foot.

Ringworm Case Study

Category	Details
Type	Fungal disease; not caused by worms.
Spread	Direct skin contact, shared towels/clothes, contaminated surfaces.
Symptoms	Circular red rash, itchy skin, red patches.
Prevention	Good hygiene, do not share towels, keep skin dry, treat infected areas.

17. Covid-19 and Ebola Case Studies

Disease	Pathogen type	Spread	Symptoms/prevention
Covid-19	Virus	Airborne droplets, direct contact, contaminated surfaces.	Symptoms: fever, cough, tiredness, breathing problems, loss of taste/smell. Prevention: vaccination, hand washing, ventilation, isolation.
Ebola	Virus	Body fluids such as blood, vomit, saliva and infected fluids.	Symptoms: fever, weakness, internal bleeding, severe illness, organ failure in serious cases.

18. Body Defences and Immunity

Defence	Role
Skin	Physical barrier preventing pathogens entering.
Mucus	Traps pathogens in airways/nose.
Cilia	Tiny hairs that move mucus away from lungs.
Stomach acid	Hydrochloric acid kills pathogens swallowed with food.

White Blood Cells

White blood cell	Function
Phagocytes	Engulf and digest pathogens. This is phagocytosis.
Lymphocytes	Produce antibodies and memory cells.

Antigens and Antibodies

Term	Meaning
Antigen	Molecule on the surface of a pathogen.
Antibody	Protein made by lymphocytes that binds to specific antigens.

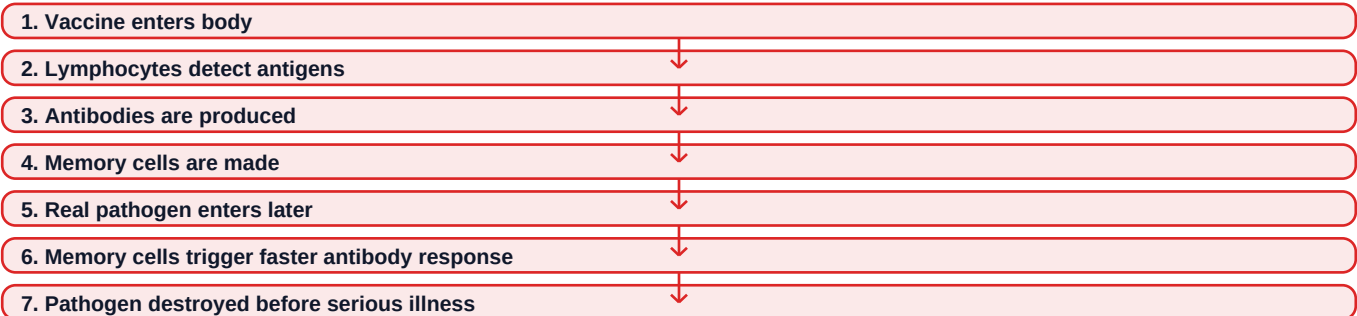
Common confusion

- Pathogens have antigens.
- Lymphocytes make antibodies.
- Lymphocytes do NOT make antigens.

19. Vaccination, Medicines and Penicillin

Vaccination

Vaccination introduces weakened, inactive or harmless pathogen material into the body to stimulate the immune system.



Medicines and Treatments

Treatment	Used for	Important detail
Antibiotics	Bacterial infections	Kill bacteria or stop bacterial growth; do not work on viruses.
Antivirals	Some viral infections	Slow virus reproduction; do not usually directly kill viruses.
Painkillers	Symptoms	Reduce pain/fever but do not kill pathogens.
Antiseptics	Skin/living tissue	Kill microbes on living tissue.
Disinfectants	Surfaces	Kill microbes on non-living surfaces; usually stronger than antiseptics.

Penicillin

Penicillin was discovered by Alexander Fleming and was the first antibiotic. It kills bacteria, not viruses.

20. Balanced Diet and Deficiency Diseases

A balanced diet contains the right amounts of carbohydrates, proteins, fats/lipids, vitamins, minerals, fibre and water.

Nutrient	Function	Examples/sources
Carbohydrates	Main source of energy.	Bread, pasta, rice, potatoes, sugar.
Proteins	Growth and repair of tissues.	Meat, fish, eggs, beans, lentils.
Fats/lipids	Energy storage, insulation, cell membranes.	Oils, butter, nuts, dairy.
Fibre	Helps digestion and prevents constipation.	Fruit, vegetables, whole grains.
Water	Hydration, transport, chemical reactions, temperature control.	Water, drinks, food.
Vitamins	Healthy body functions.	Vitamin C for tissues; Vitamin D for bones.
Minerals	Essential body processes.	Iron for haemoglobin; calcium for bones/teeth.

Deficiency Diseases

Disease	Deficiency	Symptoms	Why it matters
Scurvy	Vitamin C	Bleeding gums, tiredness, poor wound healing, weakness.	Vitamin C helps maintain healthy tissues.
Rickets	Vitamin D / calcium	Weak bones, bone pain, bowed legs.	Vitamin D helps calcium absorption and strong bones.
Anaemia	Iron	Tiredness, weakness, dizziness, pale skin.	Iron is needed for haemoglobin to carry oxygen.

21. Inherited Diseases and Alleles

Inherited diseases are passed from parents to offspring through genes. They are non-communicable and are not caused by pathogens.

Example inherited disease	Key point
Cystic fibrosis	Genetic disorder often used as a recessive inheritance example.
Sickle cell disease	Inherited condition affecting red blood cells.
Huntington's disease	Inherited condition often used as a dominant inheritance example.

Genes, Chromosomes and Alleles

Term	Meaning
DNA	Carries genetic instructions.
Chromosome	Long structure made of DNA. Humans have 23 pairs.
Gene	Section of DNA controlling a characteristic.
Allele	Different version of a gene.
Dominant allele	Only one copy needed to be expressed.
Recessive allele	Two copies needed to be expressed.
Genotype	Trait shown
AA	Dominant trait shown
Aa	Dominant trait shown
aa	Recessive trait shown

Important inheritance warnings

- Dominant does not mean better.
- Recessive does not mean weaker.
- Dominant/recessive only describe how alleles are expressed.

SECTION 3: EVOLUTION

Variation, natural selection, Darwin, moths, fossils and human evolution

22. Evolution and Variation

Evolution is the gradual change in species over many generations. Variation means differences between individuals.

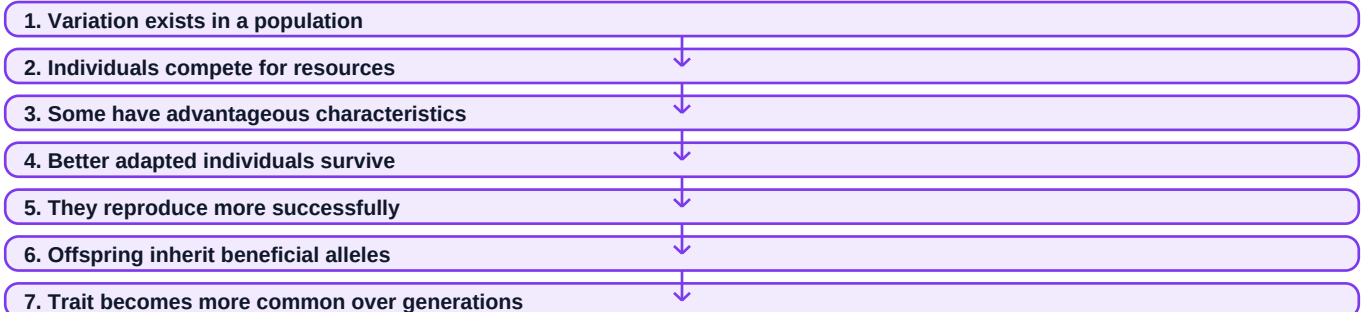
Cause of variation	Explanation
Inherited genes	Different alleles inherited from parents.
Mutations	Random changes in DNA that can create new alleles.
Environment	Differences caused by surroundings, diet, climate or lifestyle.

Mutation

A mutation is a random change in DNA. Mutations can have no effect, be harmful, or be beneficial. Beneficial mutations can create advantageous alleles that may spread through natural selection.

23. Natural Selection

Natural selection is the process where organisms with advantageous characteristics survive and reproduce more successfully.



Key exam words

- variation
- competition
- adaptation
- survival
- reproduction
- inheritance
- alleles
- generations

24. Darwin's Finches

Charles Darwin studied finches on the Galapagos Islands. Different islands had different food sources, and finches had different beak shapes.

Observation	Natural selection explanation
Different food sources on different islands	Different beak shapes were useful in different environments.
Strong beaks	Useful for cracking seeds.
Thin beaks	Useful for catching insects.
Best-adapted finches survived	They reproduced more successfully.
Offspring inherited beak traits	Over generations, populations became different.

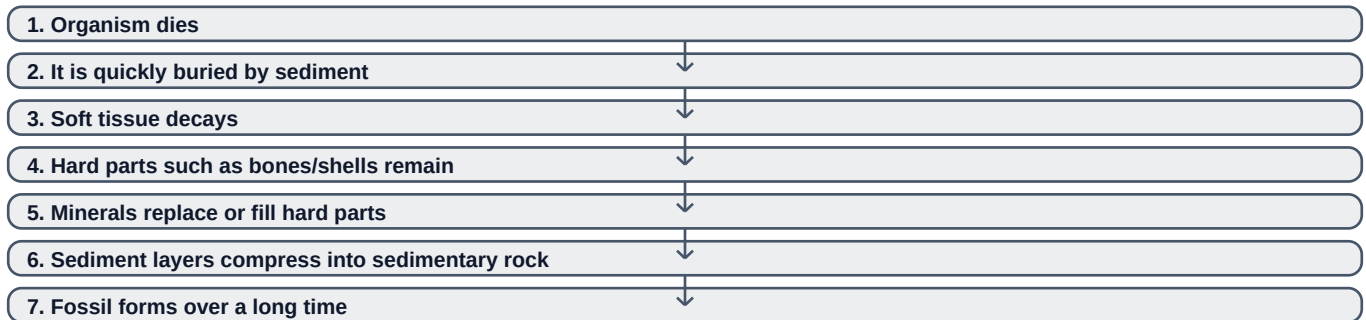
25. Peppered Moths

Time period	Environment	Which moth survived better?	Reason
Before Industrial Revolution	Trees were lighter	Light moths	Better camouflaged; fewer eaten by birds.
During Industrial Revolution	Soot darkened trees	Dark moths	Better camouflaged; survived and reproduced more.
After pollution decreased	Trees became lighter again	Light moths increased; dark moths decreased	Dark moths were easier for predators to see.

This is natural selection because the environment changed which trait was advantageous.

26. Fossils

A fossil is the preserved remains or trace of an ancient organism. Fossils provide evidence for evolution and extinct organisms.



Why Fossils Are Useful

- Show organisms that lived in the past.
- Provide evidence for evolution.
- Show extinct species.
- Show how environments changed over time.

Limitations of the Fossil Record

- Many organisms decay before fossilising.
- Soft-bodied organisms rarely fossilise.
- Fossils can be destroyed by erosion or geological activity.

- Many fossils have not been found yet.

27. Extinction and Human Evolution

Extinction means a species has completely died out.

Causes of extinction
Environmental change
Disease
New predators
Competition
Habitat loss
Catastrophic events

Human Evolution

Modern humans are called **Homo sapiens**. Humans evolved from earlier hominin species over millions of years. Evidence includes fossils, skeleton comparisons, DNA similarities and tools.

Classification level	Human classification
Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Primates
Family	Hominidae
Genus	Homo
Species	sapiens

Geological Periods Mentioned

Period	Key idea
Cambrian	Many early animal groups appeared; often linked to the Cambrian explosion.
Carboniferous	Large forests; coal-forming period; many insects and amphibians.
Triassic	Early dinosaurs appeared.
Jurassic	Dinosaurs became dominant.
Cretaceous	Many dinosaurs including Tyrannosaurus rex; ended with mass extinction.

28. Classification and Ecology Keywords

Keyword	Definition
Species	A group of organisms that can reproduce together and produce fertile offspring.
Habitat	The place where an organism lives.
Adaptation	A characteristic that helps an organism survive in its environment.
Producer	Organism that makes its own food, usually by photosynthesis.

Keyword	Definition
Consumer	Organism that eats other organisms.
Herbivore	Eats plants.
Carnivore	Eats animals.
Omnivore	Eats both plants and animals.
Decomposer	Breaks down dead material; examples include bacteria and fungi.

29. Data and Practical Skills

Skill/term	Meaning	Example
Independent variable	What you change.	Distance from lamp.
Dependent variable	What you measure.	Oxygen bubbles per minute.
Control variable	What you keep the same.	Temperature, pondweed species, time counted.
Positive correlation	Both variables increase together.	Light intensity up, bubbles up.
Negative correlation	One variable increases as the other decreases.	Distance up, bubbles down.
Anomaly	A result that does not fit the pattern.	One bubble count much lower than expected.

Trend question rule

- If asked to describe a trend, say what the data shows.
- Do not explain why unless the question asks you to explain.
- Example: As distance from the lamp decreases, oxygen bubbles per minute increase.

30. Key Definitions to Memorise

Term	Definition
Botany	Study of plants.
Cell wall	Cellulose structure that supports plant cells.
Chloroplast	Organelle where photosynthesis occurs.
Chlorophyll	Green pigment that absorbs light energy.
Vacuole	Contains cell sap and maintains turgor pressure.
Osmosis	Movement of water through a partially permeable membrane from high water concentration to low water concentration.
Diffusion	Movement of particles from high concentration to low concentration.
Transpiration	Loss of water vapour through stomata.
Translocation	Movement of sugars through phloem.
Photosynthesis	Process where plants use light energy to make glucose.
Pollination	Transfer of pollen from anther to stigma.
Fertilisation	Fusion of male and female nuclei.
Gamete	Sex cell.

Term	Definition
Zygote	Cell formed after fertilisation.
Pathogen	Microorganism that causes disease.
Communicable disease	Disease that can spread between organisms.
Non-communicable disease	Disease that cannot spread between organisms.
Antigen	Molecule on pathogen surface.
Antibody	Protein made by lymphocytes that binds to antigens.
Vaccination	Giving harmless pathogen material to stimulate immunity.
Antibiotic	Medicine that kills bacteria.
Inherited disease	Disease passed through genes.
Gene	Section of DNA controlling a characteristic.
Allele	Different version of a gene.
Natural selection	Better adapted organisms survive and reproduce more successfully.
Fossil	Preserved remains or trace of an ancient organism.
Extinction	When a species completely dies out.

31. Common Exam Mistakes to Avoid

Mistake	Correct version
Chloroplasts absorb sunlight.	Chlorophyll absorbs light energy; chloroplasts are where photosynthesis occurs.
Chlorophyll is a reactant.	Chlorophyll is a pigment/condition needed for light absorption.
Gas exchange happens by osmosis.	Gas exchange happens by diffusion; osmosis is water only.
Pollination is when nuclei join.	Pollination is transfer of pollen; fertilisation is fusion of nuclei.
Lymphocytes produce antigens.	Pathogens have antigens; lymphocytes produce antibodies.
Viruses are living because they reproduce.	Viruses cannot reproduce independently and are not considered living.
Seeds disperse to find food.	Seeds disperse to reduce competition for light, water, space and minerals.
Organisms choose to adapt.	Organisms with inherited advantageous traits survive and reproduce more.
Dominant means better.	Dominant only means one allele is needed to express the trait.

32. Final Monday Priority Checklist

Memorise these perfectly

- Plant cell structures and functions
- Specialised plant cells: palisade, root hair, xylem, phloem, guard cells
- Photosynthesis word and symbol equations
- Uses of glucose
- Stomata, diffusion and transpiration
- Pondweed variables: independent, dependent, control
- Flower parts and functions
- Pollination vs fertilisation
- Seed dispersal methods
- Pathogen definition and four types
- Bacteria structure
- Virus structure: capsid + genetic material
- Case studies: Salmonella, Covid-19, Ebola, Ringworm
- Body defences and white blood cells
- Antibiotics vs antivirals vs antiseptics vs disinfectants
- Vaccination process
- Balanced diet and deficiency diseases
- Genes, alleles, dominant and recessive
- Natural selection steps
- Darwin's finches and peppered moths
- Fossil formation
- Graph/data skills